Vardhan Dongre

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EDUCATION

University of Illinois, *Urbana-Champaign, IL*
Master of Science in Computer Science, GPA: 3.83 / 4.00 — *Aug. 2020 – May. 2022*

University of Illinois, *Urbana-Champaign, IL*
Master of Science in Civil Engineering, GPA: 3.76 / 4.00 — *Aug. 2018 – Aug. 2020*

National Institute of Technology, *Bhopal, India*
Bachelors of Technology in Civil Engineering, GPA: 8.92 / 10.00 — *Jul. 2013 – Apr. 2017*
Awarded Medal of Excellence for Academic Performance

WORK EXPERIENCE

University of Illinois, *Champaign, IL*
Course Director, Data Curation MCS-DS Coursera — *Aug. 2021 – Present.*

- Implemented PII Encryption strategies and Data anonymization using Apache Spark and IBM Differential Privacy API
- Developed learning modules on Apache Spark, Delta Lake Tables and warehousing for cloud architecture development
- Assisted in developing autograding platform for Coursera-Illinois online degree platform
- Managing a staff of 18 graduate students to work on maintaining data curation processes

No. of students for xxx students.

Brunswick Corporation I-Jet Lab, *Champaign, IL*
Computer Vision & Deep Learning Engineer Intern — *May. 2021 – Aug. 2021*

Engineered

- Developed a scalable Azure ML pipeline for automated data labeling & augmentation for perception algorithms
- Proposed stereo vision based wave pattern recognition and early warning system [for what purpose]
- Designed a gated multi-headed perception architecture for multimodal representation learning from LIDAR, camera & IR
- Developed a multi-modal semantic segmentation framework to be deployed on prototype perception system of autonomous boats
- Technologies worked on: Sensor Fusion, LIDAR and point cloud learning frameworks, Stereo Vision, ROS, RTMaps

Illinois Solar Decathlon, *Champaign, IL*
Structural Engineer & IOT Developer, Build Team — *Aug. 2018 – May. 2019*

- Developed structural designs for Gravity framing, lateral load resisting system & connection details of a modular building
- Developed IOT home automation software system for controlling HVAC, water supply and power for the smart home
- Developed new design documentation which increased the revenue savings by $130,000

Replaced conventional concrete prefab cold formed steel board

RESEARCH & ACADEMIC PROJECTS *

GRAPHICS & COMPUTER VISION

Efficient Pose and Velocity estimation using visual-inertial attention-based sensor fusion
University of Illinois, Urbana-Champaign — *Jan. 2022 – Apr. 2022*

- Developed a novel encoder – fusion – regressor based architecture with ResNet / TCN backbones for estimating positions and orientations of moving subjects
- Proposed attention-based sensor fusion for visual and inertial inputs to generate robust representations
- Achieved improvements on existing VIO methods in terms of ATE and RTE improvement by 2%

IBL based rendering synthetic objects in real scenes
University of Illinois, Urbana-Champaign — *Sept. 2020 – Oct. 2020*

- Developed an Image-based light model from geometric model of the real scene using omni-directional radiance map captured from light probes
- Generated HDR images by merging sequence of LDR images and estimating the camera response function
- Integrated synthetic objects in real photographs and videos seamlessly

Deep Learning Assisted Camera Pose Estimation for Visual Odometry [Code] [Report] ,
python, pytorch — *Aug. 2020 – Dec. 2020*

- Developed pose estimation module using classical SIFT and RANSAC algorithms for KITTI dataset
- Developed a self-supervised model using synthetic data and adaptive homographies that achieved similar results to classical module

Detecting Deepfake Videos by Exploiting Spatio-Temporal Information [PPT] [Report],
python, pytorch — *Aug. 2020 – Dec. 2020*

- Designed 3D CNN based ResNets architecture for spatio temporal representation learning from videos for detecting deepfakes
- Achieved significant improvements over existing fingerprinting and Low resolution detection techniques
- Achieved a 0.92 F1 score outperforming multiple baseline detectors

Cloud Detection and Masking on NASA's Terra MODIS satellite images using Deep Learning [Code] [Viz],
pytorch, sckit-learn, fastai *Feb. 2020 – Mar. 2020*

- Developed binary classification and image segmentation approaches as baseline approaches for cloud detection in satellite imagery
- Developed U-Net based novel architecture for NASA's MODIS cloud mask product
- Model achieved 0.95 F1 score and was winning entry in NCSA-NVIDIA AI Hackathon

AUTONOMOUS SYSTEMS

Design of Autonomous Driving System and Impact on Public Policy ,
python, pytorch *Jan. 2021 – May. 2021*

- Proposed UI/UX for in-car interactive system for autonomous vehicles to handle critical, sub critical and non-critical tasks
- Proposed personalization features for driver assistance and its impact of existing policies of DOT

Performance Predictors for Meta-Learning and AutoML [Link],
python, tensorflow, keras *Aug. 2019 – Dec. 2019*

- Developed a performance predictor model to automate the model selection process
- Developed a seq2seq model using tf and keras that takes as input model architecture and hyperparameters and predicts its final performance with 97% accuracy.

SIGNAL PROCESSING / ADVERSARIAL ML

Detecting deepfake audio for preventing attacks on ASV systems [PPT] [Report],
python, pytorch *Aug. 2021 – Dec. 2021*

- Implemented end2end residual framework as baseline for classifying frontend audio features like Constant Q Cepstral coefficients, Inverted Melfrequency cepstral coefficients and Linear frequency Cepstral coefficients
- Proposed a novel architecture based on ResNet34 and Squeeze Excitation blocks trained on fused features obtained using attentional feature fusion mechanism that outperforms baselines

HIGH PERFORMANCE / HETEROGENEOUS COMPUTING

Distributed Training of Deep Learning Models on GPU clusters,
pytorch, CUDA, OpenMP, MPI, Globus, Horovod *Aug. 2019 – Oct. 2019*

- Developed distributed deep learning training framework for synchronous and asynchronous training of the network for video captioning.
- Developed program to check dropout test using heuristics and Monte Carlo simulation
- Wrote PBS using aprun utility for running hybrid jobs with OpenMP/MPI
- Implemented data parallelism in pytorch for distributed training of Reinforcement Learning models on Atari games using horovod.

Deep Learning based Structural Topology Optimization
University of Illinois,, Urbana-Champaign *Jan. 2019 – May. 2019*

- Designed and performed numerical experiments on non-convex optimization problems
- Developed and curated image dataset for optimized topolgies for standard structural domains
- Developed a U-Net based architecture for segmentation and predicting topologies from partial inputs from PDE solvers

* Refer website/Github for other works

SKILLS

Machine Learning: XGBoost, PCA, Clustering, GLM, Ensembles, E-M, GMM, HMM
Deep Learning: Segmentation, NLP, Transfer Learning, CNN, VAE, Sequence Models, DDQN, A3C
Programming Languages*: Python(4), Shell Scripting(4), SQL(3), R(3), C++(3), PHP(3), Java(2), CUDA(2)
Tools & Libraries: Scikit-learn, Pytorch, Tensorflow, Pandas, Matplotlib, NumPy, Jupyter Notebook, Google colab
Modeling & Simulation: Unreal Engine, Blender, AUTOCAD, REVIT, BIM360
Github/Web : Github/Web
* Proficiency on a scale of 1 (lowest) to 4 (highest)

CERTIFICATION / AWARDS

Google Cloud Machine Learning Engineer: - Anticipated June 2022
Autodesk Certified AutoCAD and Fusion 360 modeler:
Bentley Certified STAAD Pro and OpenRoads designer:
Department Medal of Excellence (National Institute of Technology, Bhopal): Awarded Silver Medal for achieving rank 2 in class of 2017